Standard + Essential
Question:
How do you find the coordinates of points on a unit circle when the angle is from special right triangles?

Questions:
How can we convert from radians to degrees and from degrees to radians?

Do Now:

a) $56°$: $1 = \frac{\pi}{180}$; $56° = \frac{\pi}{180}(56)$; $56° = \frac{56\pi}{180} = \boxed{\frac{14\pi}{45}}$

b) $\pi/8$: $1 = \frac{180}{\pi}$; $\frac{\pi}{8} = \frac{180}{\pi} \cdot \frac{\pi}{8}$; $\frac{\pi}{8} = \frac{180}{8}$; $\boxed{\pi/8 = 22.5°}$

c) $256°$: $1 = \frac{\pi}{180}$; $256° = \frac{\pi}{180}(256)$; $256° = \frac{256\pi}{180}$; $\boxed{256° = \frac{64\pi}{45}}$

d) $\frac{12\pi}{10}$: $1 = \frac{180}{\pi}$; $\frac{12\pi}{10} = \frac{180}{\pi}\left(\frac{12\pi}{10}\right)$; $\boxed{\frac{12\pi}{10} = 216°}$

Notes:

1) A, B, C; 1; 0.3
$\sin A = \frac{0.3}{1}$
$\sin^{-1}(\sin A) = \sin^{-1}\left(\frac{0.3}{1}\right)$
$\boxed{A = 17.46°}$

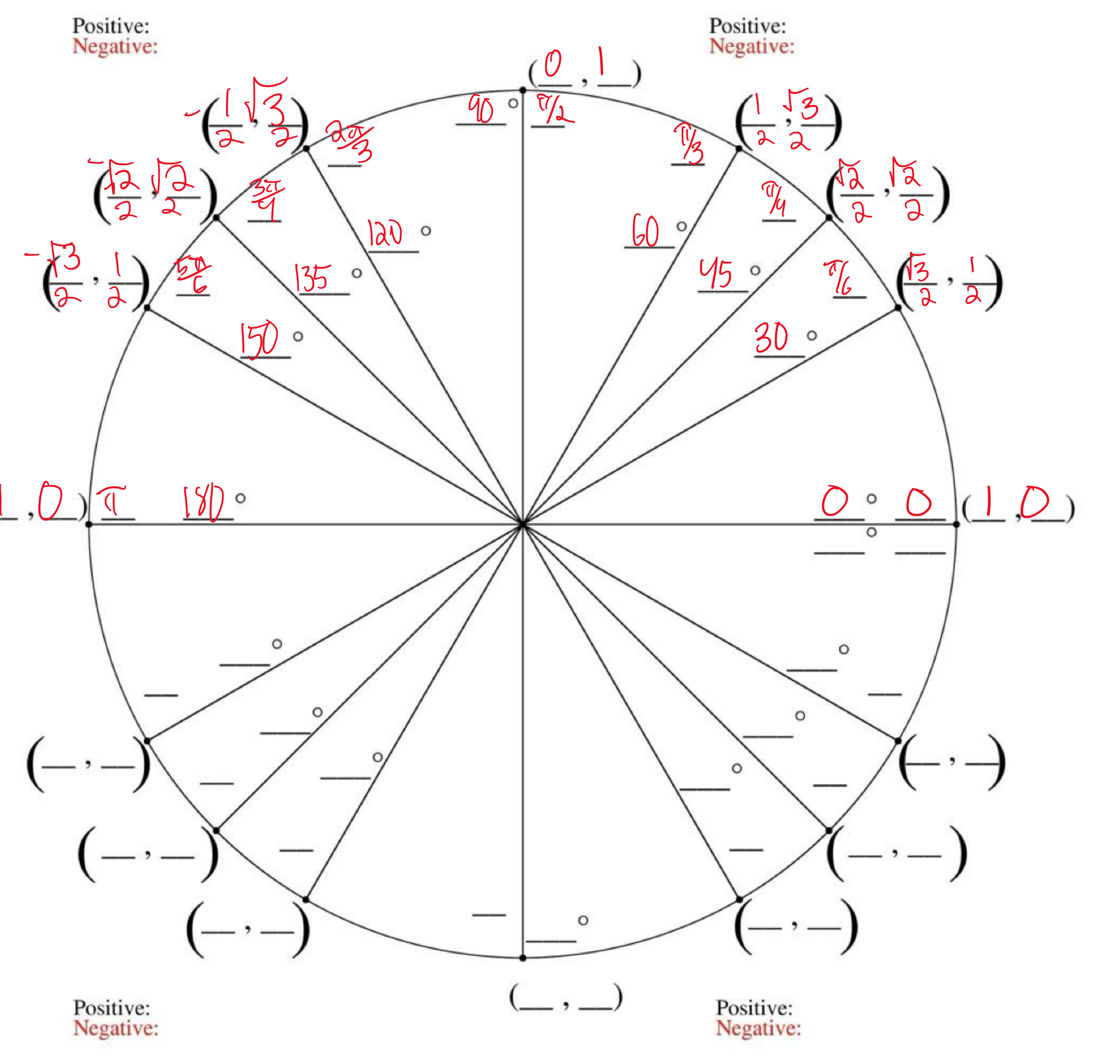
2) halfway (45°, 135°); 1; -1; 90; 180; 270; 360; initial; bottom
θ | y
degrees | height

3) $Q = (\cos(110), \sin(110))$
$P = (\cos(50), \sin(50))$
1; y; x

4) b) $= 57.29°$
c) $= 180°$
d) $= 90°$
e) $= 45°$
f) $= 60°$
g) $= 30°$

Summary:

Fill In The Unit Circle



9.1.6 What do I know about a unit circle?

Building a Unit Circle

In this lesson you will further develop your understanding of the unit circle and how useful it can be. By the end of the lesson, you should be able to answer the questions below.
What can the unit circle help me understand about an angle?
What can angles in the first quadrant tell me about angles in other quadrants?

9-72. There are some angles for which you know the *exact values* of sine and cosine. In other words, you can write the exact sine and cosine without using a calculator. Work with your team to list as many such angles (expressed in radians) as you can.

9-73. Now you will build a unit circle. Obtain the Lesson 9.1.6 Resource Page. There are points shown along the circle for angles of $\frac{\pi}{12}$, $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{5\pi}{12}$, $\frac{7\pi}{12}$, $\frac{2\pi}{3}$, $\frac{3\pi}{4}$, $\frac{5\pi}{6}$, and $\frac{11\pi}{12}$, starting from the positive x-axis.

a. Label the exact coordinates for three of the points shown in the *first quadrant.*

$30° \to \frac{\pi}{6} \to \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
$45° \to \frac{\pi}{4} \to \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
$60° \to \frac{\pi}{3} \to \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

b. Mark *all* other points in the unit circle for which you know the *exact* coordinates. Not all of the points are shown. Label each point with its angle of rotation (in radians) and its coordinates.

c. If you have not done so already, label each point with its corresponding angle measure in radians.

9-74. Draw a new unit circle, label a point that corresponds to a rotation of $\frac{\pi}{12}$, and put your calculator in radian mode.

a. What are the coordinates of this point, correct to two decimal places?

1; θ; $\frac{\pi}{12}$
$(\cos\theta, \sin\theta)$
$\left(\cos\frac{\pi}{12}, \sin\frac{\pi}{12}\right)$
$(0.97, 0.26)$

emphasize reference angles

b. Use the information you found in part (a) to determine each of the following values: (Hint: Drawing each angle on the unit circle will be helpful.)

i. $\sin\left(\frac{11\pi}{12}\right)$ — $\sin\frac{11\pi}{12} = 0.26$; Q 2

ii. $\cos\left(\frac{13\pi}{12}\right)$ — $\cos\frac{13\pi}{12} = -0.97$; Q 3

iii. **Challenge:** $\cos\left(\frac{7\pi}{12}\right)$ — $\cos\frac{7\pi}{12} = -0.26$; Q2

height + width switched
(rotation of angle not reflection)

9-75. For angle α in the first quadrant, $\cos(\alpha) = \frac{8}{17}$. Use this information to calculate each of the following values without using a calculator. Be prepared to share your strategies with the class. Note that the symbol α is the Greek letter "alpha" and, like θ, can represent the measure of an angle.

a. $\sin(\alpha) = \frac{15}{17}$

17; 15; 8
$8^2 + b^2 = 17^2$
$64 + b^2 = 289$
$-64 \qquad -64$
$\sqrt{b^2} = \sqrt{225}$
$b = 15$

homework if no time

b. $\sin(\pi + \alpha)$ $\quad -\frac{15}{17}$

c. $\cos(2\pi - \alpha)$ $\quad \frac{8}{17}$

independent if there is

1) Do Now (10 mins)
- Example given on board 3 mins
- Student work 5 mins
- 2 mins exemplar for 1 problem

2) Notes/Review (30 mins)
- Students grouped and moved to boards
- Timed responses with review after
- emphasize
  - $2\pi = 360$
  - use a table when graphing
  - coordinates on unit circle are always $\cos\theta$, $\sin\theta$
  - arcsin cancels out sin
  - arcsin = angle, sin = side

- how can we find angles on triangles given the sides?
- What ratio uses the sides given?
- How can we remove the operation?
- How do we normally graph lines?
- What do we know about sin?
- What are the important degrees we need?
- Did you graph the triangles yet?
- What do the x and y refer to?
- How many radians are in a circle?
- Can we convert like in the do now?

3) - Use unit circle given
- Students start in Q1
- goal - fill out as much as possible
- emphasize
  - creating the slope triangles
  - convert to radians for exact measurements
  - referring to special triangles
  - reference angles allow coordinates to change signs to move quadrants
- how can we go from degrees to radians?
- do we know the angle measure already?
- how do we find coordinates?
- how can we use special triangles to help solve this?

4) - Students work w/ group to solve
- Give time to focus on part A, review whole class w/ aid call after
- Part b done w/ pair
- challenge completed alone

emphasize
- reference angles
- rotations vs reflections
- sign change w/ quadrant change.

- how do signs change as the quadrants do?
- What are the reference angles?
- Does the coordinate change as the angle moves?